20	P(OOO) $\frac{6}{9}\times\frac{5}{8}\times\frac{4}{7}=\frac{120}{504}$ P(OEE) $\frac{6}{9}\times\frac{3}{8}\times\frac{2}{7}=\frac{36}{504}$ P(EOE) $\frac{3}{9}\times\frac{6}{8}\times\frac{2}{7}=\frac{36}{504}$ P(EEO) $\frac{3}{9}\times\frac{2}{8}\times\frac{3}{7}=\frac{36}{504}$ P(odd) $\frac{120}{504}+\frac{36}{504}+\frac{36}{504}+\frac{36}{504}$ P(EEE) $\frac{3}{9}\times\frac{2}{8}\times\frac{1}{7}=\frac{6}{504}$ P(EOO) $\frac{3}{9}\times\frac{6}{8}\times\frac{5}{7}=\frac{90}{504}$ P(OEO) $\frac{6}{9}\times\frac{3}{8}\times\frac{5}{7}=\frac{90}{504}$ P(OOE) $\frac{6}{9}\times\frac{5}{8}\times\frac{3}{7}=\frac{90}{504}$ P(odd) $1-\left(\frac{6}{504}+\frac{90}{504}+\frac{90}{504}+\frac{90}{504}\right)$	$\frac{228}{504}$	P1 P1 P1 P1 A1 **OR** P1 P1 P1 P1 A1	for a “second choice” denominator of 8 for one correct product of 3 probabilities for all four correct products of 3 probabilities for a fully correct process to find the probability of the sum being odd oe for a “second choice” denominator of 8 for one correct product of 3 probabilities (method to use 1 – p(odd)) for all four correct products of 3 probabilities for a fully correct process to find the probability of the sum being odd oe

Documented scheme clarification

1MA1_MOCK_SET_03_3H_Q20

Original source

scheme page 14, SHA256
7aa2e41b345577b3ec27c4f13bcc03c9fd284f3ea79baad14a4e26f52128b932

Original wording

The first method prints 3/9 x 2/8 x 3/7 for P(EEO), with result 36/504.

Correction / interpretation

For EEO, two even counters have been removed and all six odd counters remain, so use 3/9 x 2/8 x 6/7 = 36/504. The last numerator 3 in the source product is a misprint. The final probability remains 228/504 = 19/42, also confirmed by the complementary method.

Reason

Without replacement, the remaining odd count is six after two even draws.

Sol authority

beb4f8a7443797f377fa651550d12b0da38f25498812f76de1ca740f8556fcfa